

Degree: B.Tech, Semester: Third-Course work
MID-SEMESTER EXAMINATION, September 2023

Course Code: ECMTC04

Course Title: Mathematics for Signal Processing and Communication

Duration: 1 hour 30 minutes

Max. Marks: 25

Note: Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1(a)	Let $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \mid a \in R, a \neq 0 \right\}$, where R is the set of real numbers. Show that G is a group under matrix multiplication.	2.5	CO1
1(b)	Prove that in a group $(G, *)$, there is only one identity element.	2.5	CO1
2(a)	For an abelian group $(G, *)$ with identity element e , consider $H = \{x \in G \mid x^2 = e\}$ (Note that, $x^2 = x * x$). Show that H is a subgroup of G .	2.5	CO1
2(b)	Consider a cyclic group $G = \{a, a^2, a^3, a^4 = e\}$ under multiplication, where e is the identity element. What is the order of G , and how many elements can be used as generators of G ?	2.5	CO1
3(a)	Define group homomorphism. If ϕ is a homomorphism from a group $(G, *)$ to group $(G', *')$ and e and e' be their respective identity elements, then prove that $\phi(x^{-1}) = [\phi(x)]^{-1} \quad \forall x \in G$.	2.5	CO1
3(b)	"Every integral domain is a field." Is the statement true? Justify your answer with an example.	2.5	CO1
4(a)	Consider the vector space $R^3(R)$. Show that $W = \{(x, y, 2z) \mid x, y, z \in R\}$ is a subspace of $R^3(R)$.	2.5	CO2
4(b)	If a, b, c are three linearly independent vectors in a vector space $V(F)$, then prove that $a + b, b + c, c + a$ are also linearly independent.	2.5	CO2
5(a)	Determine whether or not the following vectors form a basis of the vector space $R^3(R)$: $(1, 1, 2), (1, 2, 5), (5, 3, 4)$	2.5	CO2
5(b)	Let $T: R^3(R) \rightarrow R^2(R)$ is a transformation defined as $T(a, b, c) = (a, b) \quad \forall (a, b, c) \in R^3$. Prove that T is a linear transformation. Also, find the kernel of T .	2.5	CO2